

Risk Across Tails and Timescales

Does tail dependence change with the investment horizon,
and what does ignoring it do to measured risk?

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Summary

We study a five-asset portfolio (US equities, Nasdaq, long Treasuries, gold and high-yield credit) on daily data from 12 April 2007 to 30 June 2026 (4,835 trading days). We use the maximal overlap discrete wavelet transform (MODWT) to look at the same returns at horizons from one day to more than three months, measure how often assets fall into their worst 5% together at each horizon, and then test what this does to Value-at-Risk (VaR) in a rolling out-of-sample backtest.

- **Does tail dependence change with the horizon?** Yes, but not evenly. Equities and high-yield credit crash together more as the horizon grows: their lower-tail co-exceedance rises from 0.53 at one day to 0.77 beyond 64 days, and the 90% interval for the change (+0.10 to +0.37) excludes zero. The equity-Treasury hedge weakens over long horizons (0.05 to 0.22), while gold behaves the other way (0.14 to 0.07). For the portfolio's risky sleeve against its hedge sleeve, co-exceedance rises from 0.06 to 0.19, though the interval is too wide to rule out no change.
- **What does ignoring it do?** A copula fitted to daily data, used for a 20-day horizon, understates 20-day VaR by 5.0% on average compared with a copula fitted to the matching horizon, and by more than 10.9% on one day in ten. This compares model forecasts on the same dates rather than realised losses, and the two models cannot be statistically separated in backtesting ($p = 0.32$). The common $\sqrt{h}$ rule hides this, because it overstates volatility by even more (8.7% on average).
- **Recommendation.** For positions held 20 trading days or longer, stop scaling 1-day VaR by $\sqrt{h}$. Use the GARCH volatility forecast for the full horizon and a copula fitted to the matching wavelet horizon. Out of sample this was the best-scoring 20-day model, kept correct coverage and needed 4.7% less capital on average than the $\sqrt{h}$ rule (Section 4).

1 Question, data and portfolio

Risk models are usually fitted on daily returns, but many portfolios are judged over weeks or months. If the way assets crash together depends on the horizon, a daily model gets the long-horizon tail wrong even when its volatility forecasts are right. We ask whether that happens and how much it matters for measured risk.

We use five liquid ETFs that each stand for a different source of risk, with fixed weights typical of a balanced portfolio: SPY (S&P 500, 30%), QQQ (Nasdaq 100, 20%), TLT (20+ year Treasuries, 25%), GLD (gold, 15%) and HYG (high-yield credit, 10%). We call SPY, QQQ and HYG the *risky sleeve* and TLT and GLD the *hedge sleeve*. A two-sleeve view turns the question into one a portfolio manager cares about: does the hedge still pay off in a crash when you hold for longer?

Prices are daily adjusted closes. The sample starts on the first day HYG traded, so it includes the 2008 crisis, the 2011 downgrade, COVID in 2020, the 2022 rates shock and the 2025 tariff selloff. We use log returns. The prices are stored in the repository so the analysis runs without a download.

2 Method

2.1 Horizon views from the wavelet transform

The MODWT splits each return series r_t into detail series $D_1, \dots, D_J$ and a smooth S_J that add back to the original exactly, $r_t = \sum_{j=1}^J D_{j,t} + S_{J,t}$ [8]. Detail D_j holds movements that last roughly 2^j to 2^{j+1} days. We use $J = 5$, which reaches beyond 64 trading days (about a quarter), and the sym8 filter. Figure 1 shows the result for SPY and TLT.

Detail bands on their own are not holding-period returns. An investor who holds for h days sees every movement that lasts longer than about h days, not one narrow band. So we build cumulative *horizon views*:

$$V_t^{(0)} = r_t, \quad V_t^{(j)} = \sum_{k=j+1}^J D_{k,t} + S_{J,t}, \quad j = 1, \dots, J, \quad (1)$$

where view j keeps only movements longer than 2^{j+1} days. An h -day return is itself a low-pass filter

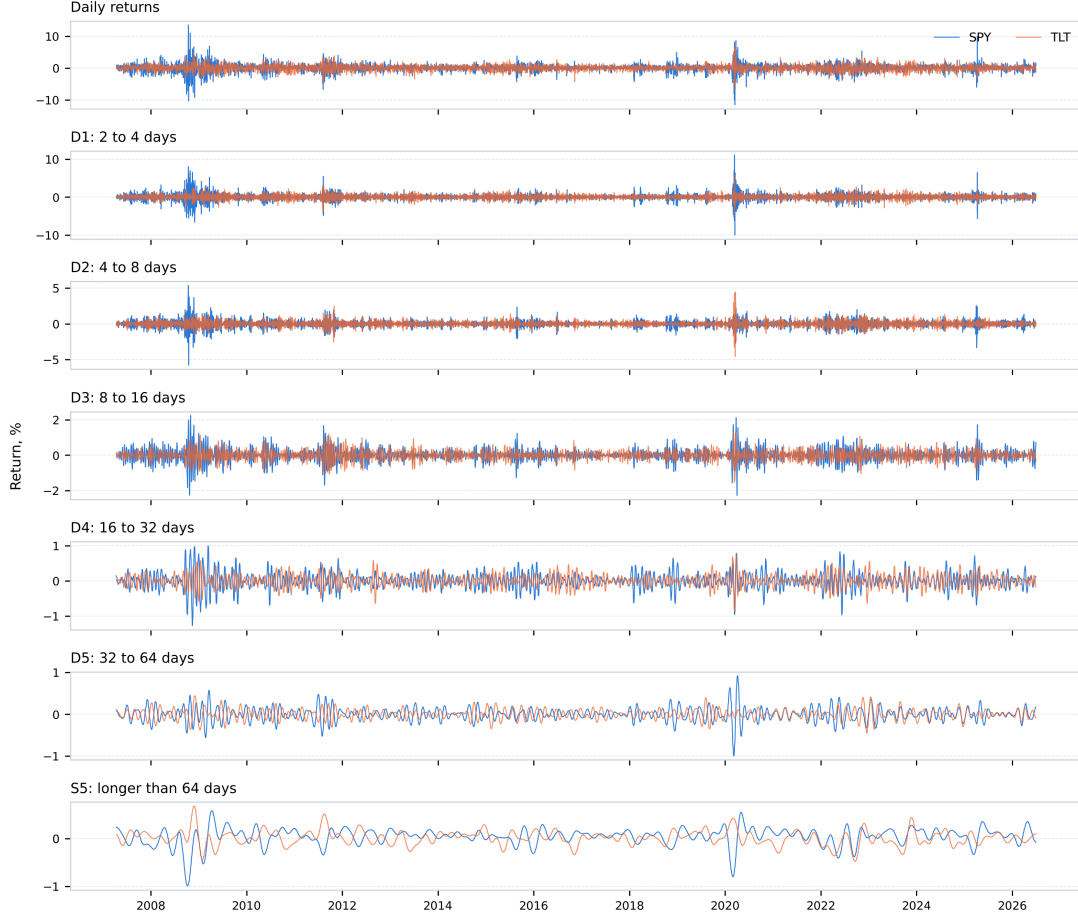


Figure 1: MODWT multiresolution analysis of SPY and TLT daily returns. The details and the smooth add back to the returns to machine precision.

of daily returns, so view j is a smooth stand-in for holding about 2^{j+1} days. Unlike non-overlapping h -day returns, it keeps every daily observation, which matters most at the longest horizons where non-overlapping data would leave only about 75 points.

Most of the variance sits at short scales: on average 79.6% of each asset’s variance is in D_1 and D_2 and only 5.2% in D_5 and S_5 (Table 1). A copula fitted to daily returns is therefore dominated by what happens over a few days.

Table 1: Share of each asset’s return variance in each MODWT scale (%).

	SPY	QQQ	TLT	GLD	HYG
D1	62.1	60.7	56.3	55.2	54.5
D2	19.5	20.2	24.1	22.6	22.9
D3	9.9	10.2	10.5	11.5	11.9
D4	3.9	4.1	4.1	5.3	4.4
D5	2.0	2.1	2.0	2.6	3.0
S5	2.5	2.7	3.0	2.8	3.4

2.2 Measuring tail dependence at each horizon

For each view we convert every series to ranks, $u_{i,t} = \text{rank}(x_{i,t})/(n+1)$. Ranks remove the margins entirely, which is what a copula needs, and they are exactly reproducible. Our main measure is the lower-tail co-exceedance at level q :

$$\hat{\lambda}_L(q) = \frac{1}{q} \Pr(U_1 \leq q, U_2 \leq q), \quad (2)$$

the share of one asset's worst- q days on which the other is also in its worst q . It equals q for independent assets and 1 for perfectly dependent ones. We use $q = 5\%$, which leaves about 240 tail days at each horizon. The upper tail is defined the same way.

Correlation alone produces some co-exceedance, so we compare each estimate with what a Gaussian copula with the same normal-scores correlation ρ would give, $\Phi_2(\Phi^{-1}(q), \Phi^{-1}(q); \rho)/q$. Co-exceedance well above that line is tail dependence that correlation does not explain.

The smooth views are strongly autocorrelated, so they contain far fewer independent observations than their length suggests. We use a moving block bootstrap with blocks of 126 days (about six months, long enough to hold a few cycles of the slowest view) and 500 replications. The same resampled days are used for every view, so we can also put an interval on the change from daily to each horizon.

2.3 Copula family

For every pair and every view we fit Gaussian, Student- t , Clayton, Gumbel and Frank copulas by maximum likelihood and compare them by BIC. The t copula wins 40 of 42 contests, so all VaR models below use a t copula. Because it has symmetric tails, its job in the VaR model is to carry the overall strength of tail dependence at each horizon. The asymmetry between lower and upper tails is reported from the empirical co-exceedances.

2.4 VaR models

All copula models share the same margins, so any difference between them comes only from the dependence structure. Each asset gets a GJR-GARCH(1,1) model with t errors [4], which captures volatility clustering and the larger effect of bad news. The standardised residuals are modelled with an empirical body and generalised Pareto tails beyond the 10% and 90% points [6]. For an h -day horizon the variance is the sum of the GARCH forecasts for days $1, \dots, h$, so volatility mean reversion is included.

We compare five models of the h -day portfolio loss at 99%:

- **Historical simulation** on overlapping h -day returns, and a **Gaussian** $\sqrt{h}$ rule: the two simple benchmarks.
- **Daily copula**, $\sqrt{h}$: 1-day VaR from the GARCH-EVT- t -copula model, scaled by $\sqrt{h}$. This is common practice.
- **Daily copula, h -day vol**: the same daily copula, with the full-horizon GARCH volatility instead of $\sqrt{h}$.
- **Horizon copula (proposed)**: the same margins, with the t copula fitted to the matching horizon view: view 1 (> 4 days) for $h = 5$, view 3 (> 16 days) for $h = 20$ and view 5 (> 64 days) for $h = 60$.

The second and third copula models differ only in volatility, the third and fourth only in dependence, so the comparisons separate the two effects. VaR and expected shortfall (ES) come from 20,000 simulated scenarios. Every model uses the same random numbers, so differences are not simulation noise.

2.5 Out-of-sample backtest

At the close of each day t we forecast the loss over days $t + 1$ to $t + h$ using only data up to t . All models are re-estimated every 21 trading days on the last 1,000 days, and the GARCH volatilities are updated every day in between. The wavelet views are computed inside each window, so the two-sided filter never sees the future. The first forecast is in March 2011, giving 3,835 forecast days per horizon.

We judge the models with the Kupiec coverage test [5], the Christoffersen independence test [2], the worst Basel traffic-light zone over any 250-day window [1], and the Fissler-Ziegel FZ0 score for VaR and ES jointly [3, 7]. FZ0 is a strictly consistent scoring rule, so lower is better and it rewards a model that is neither too high nor too low. Overlapping h -day losses are serially dependent, so coverage tests use every h -th forecast, and FZ differences are tested with a Diebold-Mariano test

Table 2: Main modelling choices and why we made them.

Choice	Reason
MODWT, not DWT	Keeps all observations at every scale and is shift invariant, so results do not depend on the start date.
sym8, level 5	Near-symmetric filter with little phase distortion; level 5 reaches past a quarter while leaving enough data at the coarsest view.
Reflection boundary	Avoids wrapping the end of the sample onto its start, which matters in the rolling windows where the last day is the one that counts.
Cumulative horizon views	An investor holding h days sees all slower movements, not one band.
Ranks, not GARCH-filtered views	GARCH is badly behaved on smooth series and gave results that changed with library versions; ranks are exact and reproducible.
$q = 5\%$ co-exceedance	Enough tail days for stable estimates at every view; still in the tail.
Block bootstrap, 126 days	Keeps the autocorrelation of the smooth views; one block length for all views makes the change versus daily testable.
GJR-GARCH- t + EVT margins	Standard for daily risk; leverage effect and fat tails matter at 99%.
Summed GARCH variance	Allows volatility to revert to its mean over the horizon, which $\sqrt{h}$ ignores.
t copula	Wins 40 of 42 BIC contests.
Rolling 1,000-day window	About four years: long enough for GARCH and EVT, short enough to adapt.
FZ0 score and DM test	Coverage tests have little power at 20 days; FZ0 uses every forecast and scores VaR and ES together.

using Newey-West standard errors.

Table 2 collects the main choices and the reason for each.

3 Results

3.1 Tail dependence across horizons

Table 3 and Figure 2 show the main result. Three things stand out.

Equity and credit crash together more at longer horizons. SPY-HYG lower-tail co-exceedance is 0.53 at one day and 0.77 beyond 64 days, well above the 0.44 a Gaussian copula with the same correlation would give. The change, +0.24 with a 90% interval of +0.10 to +0.37, is the clearest horizon effect in the data. Credit losses build over weeks as spreads widen, so a credit position that looks only partly linked to equities day by day is closely linked over a quarter.

The Treasury hedge weakens over long horizons, gold does not. On daily data SPY and TLT almost never crash together (0.05, about what independence would give). Beyond 64 days the figure is 0.22 (interval 0.01 to 0.34). All of the joint crash days at that horizon come from two episodes, early 2009 and 2022, when stocks and long bonds fell together for months. Gold moves the other way, from 0.14 to 0.07.

For the whole hedge sleeve the direction is clear but the size is uncertain. Co-exceedance of the risky and hedge sleeves rises from 0.06 at one day to 0.19 beyond 64 days. At every horizon it is above the Gaussian benchmark (0.02 at one day). The change of +0.13 has a 90% interval of -0.05 to +0.33: the longest views contain only a handful of independent episodes, and we do not claim more than the data supports. The upper tail does not rise in the same way, so the long-horizon increase is a crash effect rather than a general rise in co-movement.

Table 3: Lower-tail co-exceedance at the 5% level by horizon view. Brackets are 90% moving block bootstrap intervals.

Horizon	Risky vs hedge	Change vs daily	Gaussian	SPY-TLT	SPY-HYG	SPY-GLD
Daily	0.06 [0.04, 0.09]	–	0.02	0.05	0.53	0.14
> 4d	0.09 [0.05, 0.13]	+0.02 [+0.00, +0.07]	0.03	0.07	0.55	0.17
> 8d	0.11 [0.05, 0.17]	+0.05 [+0.00, +0.11]	0.03	0.09	0.50	0.14
> 16d	0.10 [0.06, 0.16]	+0.04 [+0.00, +0.10]	0.04	0.05	0.55	0.12
> 32d	0.15 [0.01, 0.27]	+0.09 [-0.05, +0.22]	0.04	0.08	0.63	0.06
> 64d	0.19 [0.00, 0.38]	+0.13 [-0.05, +0.33]	0.04	0.22	0.77	0.07

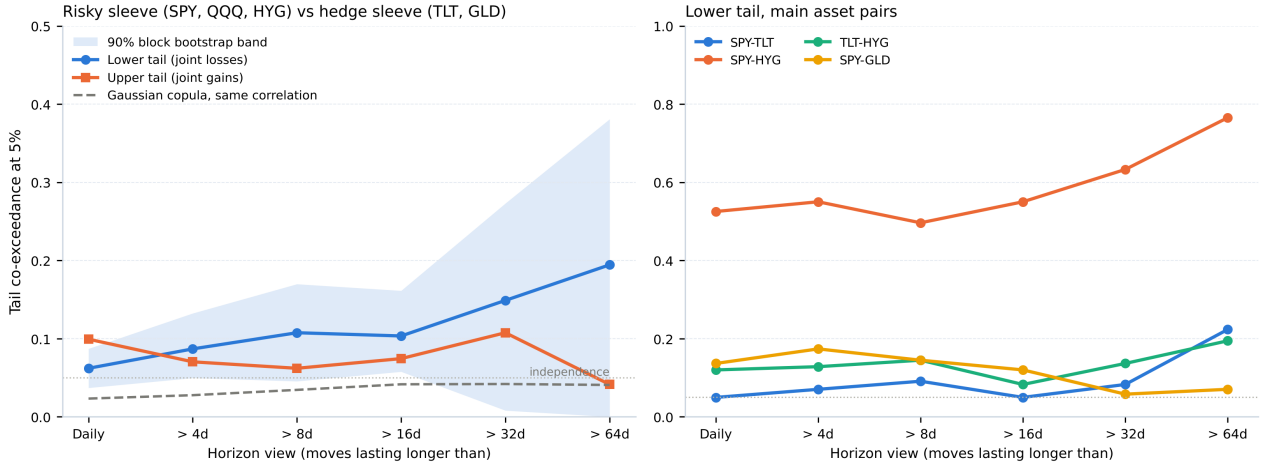


Figure 2: Left: lower and upper tail co-exceedance of the risky and hedge sleeves by horizon view, with the 90% block bootstrap band and the Gaussian copula benchmark. Right: lower-tail co-exceedance for the main asset pairs. The dotted line is independence.

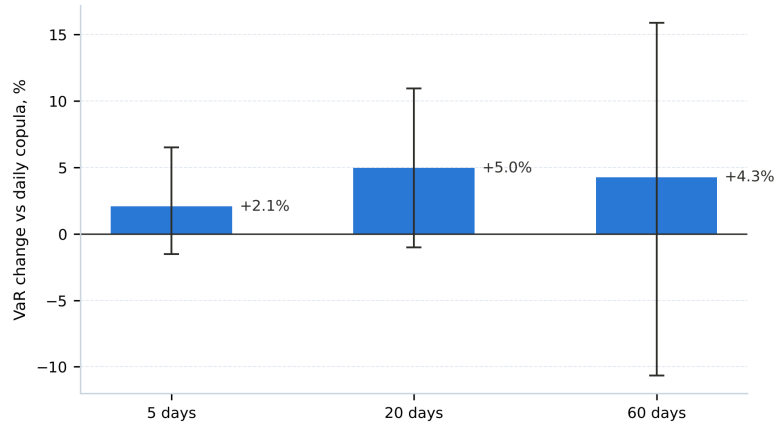


Figure 3: Average change in VaR from using the horizon copula instead of the daily copula, with the same margins and volatility. Bars are means over all forecast days; whiskers are the 10th and 90th percentiles.

3.2 What ignoring it does to measured risk

Figure 3 compares the horizon copula with the daily copula over the whole backtest. Both use the same margins and the same full-horizon volatility, so the gap is purely the effect of the dependence structure. Using horizon-matched dependence raises VaR by 2.1% at 5 days, 5.0% at 20 days and 4.3% at 60 days on average. The effect varies over time: at 20 days the 10th to 90th percentile range is -1.0% to +10.9%, and at 60 days -10.7% to +15.9%, because the coarsest view is estimated on few independent episodes.

The usual $\sqrt{h}$ rule makes the daily copula look safe. Scaling 1-day VaR by $\sqrt{h}$ gives 8.7% more VaR at 20 days than the full-horizon GARCH volatility and 18.2% more at 60 days, because high volatility tends to fall back over the horizon. That overstatement is larger than the dependence understatement, so the $\sqrt{h}$ rule ends up with enough capital for the wrong reasons. A desk that improves its volatility model but keeps daily dependence would remove the cushion and be left with too little capital.

3.3 Out-of-sample backtest

Table 4 reports the backtest. At one day the three copula models are the same forecast. They give 37 breaches against 38 expected, pass both coverage tests (Kupiec $p = 0.83$, Christoffersen $p = 0.40$) and reach the yellow zone at worst, in the 250 days after the COVID crash. Both simple benchmarks fail: the Gaussian rule has 68 breaches and historical simulation, while close on the total, bunches its 41 breaches together; both reach the red zone.

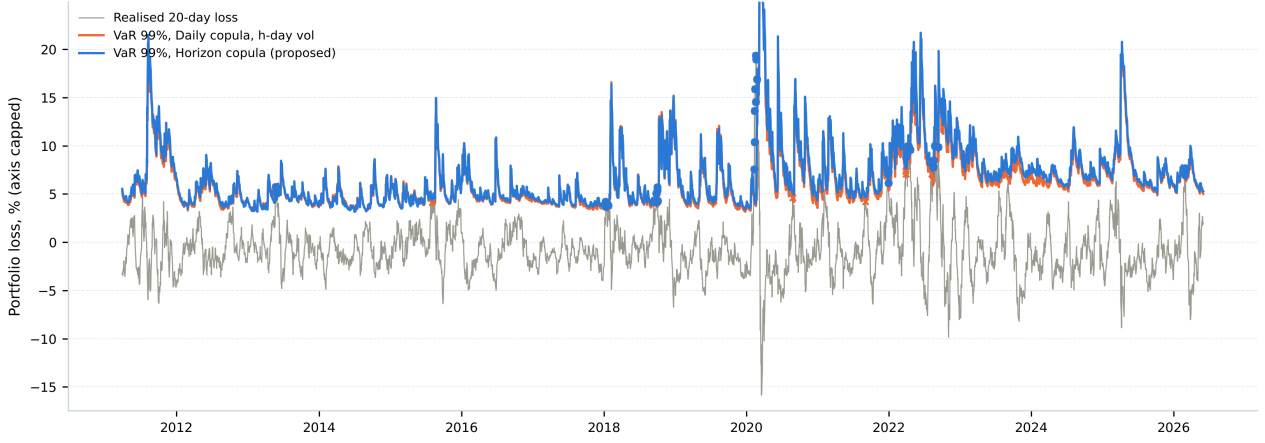


Figure 4: Realised 20-day portfolio loss against the 99% VaR of the daily copula and the horizon copula, March 2011 to June 2026. Markers show breaches. The axis is capped so the March 2020 peak does not flatten the rest.

At 20 days the horizon copula has the best FZ score of all five models, with 2 breaches in 191 non-overlapping windows against 1.9 expected. Its advantage is not statistically significant ($p = 0.32$ against the daily copula, $p = 0.25$ against $\sqrt{h}$), which is expected with fifteen years of 20-day windows. At 5 days the three copula models are within noise of each other ($p = 0.51$), which matches the small gap in Figure 3.

The 2022 rates shock is the case the horizon model is built for, with stocks and bonds falling together for months. Over its 209 forecast days the daily copula had 16 breaches at 20 days and the horizon copula 13. The $\sqrt{h}$ rule had 11, but only because it carried more capital on average over the whole sample. In the COVID crash all three had 5 breaches: the crash was too fast for any 20-day model.

Table 4: Rolling out-of-sample backtest of 99% VaR and ES. Coverage tests use non-overlapping windows. FZ is the Fissler-Ziegel score (lower is better); ΔFZ is the difference to the daily copula with its Diebold-Mariano p -value.

h	Model	Breaches / exp.	Kupiec p	Christ. p	FZ	ΔFZ (p)	Avg VaR	Basel
1d	Historical simulation	41 / 38.4	0.67	0.00	-3.612	+0.308 (0.05)	1.75%	Red
1d	Gaussian, $\sqrt{h}$	68 / 38.4	0.00	0.00	-3.381	+0.539 (0.06)	1.47%	Red
1d	All three copula models	37 / 38.4	0.83	0.40	-3.920	–	1.60%	Yellow
5d	Historical simulation	6 / 7.7	0.53	0.76	-2.686	+0.271 (0.23)	3.92%	–
5d	Gaussian, $\sqrt{h}$	10 / 7.7	0.42	0.11	-2.379	+0.578 (0.17)	3.19%	–
5d	Daily copula, $\sqrt{h}$	10 / 7.7	0.42	0.61	-2.973	-0.015 (0.20)	3.58%	–
5d	Daily copula, h-day vol	10 / 7.7	0.42	0.61	-2.957	–	3.45%	–
5d	Horizon copula (proposed)	10 / 7.7	0.42	0.61	-2.966	-0.008 (0.51)	3.52%	–
20d	Historical simulation	3 / 1.9	0.46	0.76	-1.804	+0.432 (0.18)	7.33%	–
20d	Gaussian, $\sqrt{h}$	2 / 1.9	0.95	0.84	-1.861	+0.376 (0.27)	6.03%	–
20d	Daily copula, $\sqrt{h}$	1 / 1.9	0.47	0.92	-2.233	+0.003 (0.94)	7.15%	–
20d	Daily copula, h-day vol	2 / 1.9	0.95	0.84	-2.237	–	6.49%	–
20d	Horizon copula (proposed)	2 / 1.9	0.95	0.84	-2.271	-0.034 (0.32)	6.81%	–

4 Recommendation

For positions held 20 trading days or longer, replace the $\sqrt{h}$ rule with a full-horizon GARCH volatility forecast and a copula fitted to the matching wavelet horizon view.

Why. On this portfolio the change needs 4.7% less 20-day capital on average than $\sqrt{h}$ (6.8% against 7.1% of portfolio value), keeps correct coverage, and has the best FZ score of the five models.

More importantly, it gets the right answer for the right reasons. The $\sqrt{h}$ rule overstates volatility and understates long-horizon dependence, and on average the first error more than covers the second. That does not hold on every day: when volatility is low the $\sqrt{h}$ rule understates it too, and both errors point the same way. Nor will it hold for a portfolio with more credit or with a stock-bond hedge that fails, as it did in 2022.

How to do it tomorrow. The change fits on top of an existing GARCH-copula VaR system:

1. Keep the daily GARCH-EVT margins. For the h -day horizon, sum the GARCH variance forecasts for days $1, \dots, h$ instead of multiplying the 1-day variance by h .
2. Once a month, compute the MODWT of the last four years of returns, build the horizon view that matches the holding period (view 3, moves longer than 16 days, for a 20-day horizon) and refit the copula to its ranks.
3. Simulate h -day losses from the new copula and the h -day margins.

All three steps are in `src/horizon_var.py`. The monthly refit adds a few seconds of computing time.

What to watch. Report the horizon-matched VaR next to the daily-copula VaR. When the gap widens, as it did through 2022, the portfolio's hedges are working less well over the holding period than daily data suggests, which is a signal to review the hedge rather than only to hold more capital.

5 Limitations

- The longest views contain few independent crash episodes. This is why the sleeve-level change and the 60-day capital gap have wide intervals. More history would help, but HYG only starts in 2007.
- The backtest needs 1,000 days of history, so it begins in March 2011. The 2008 crisis is used for estimation only and is never forecast out of sample.
- Coverage tests at 20 days rest on 191 non-overlapping windows, so they cannot tell good models apart. We rely on the FZ score, which uses every forecast, and still find differences that are not significant.
- The horizon copula is fitted to ranks of the smoothed returns without removing volatility clustering first. Shared volatility spikes may add to measured dependence. The daily copula is fitted to the same kind of data, so the comparison between them is fair, but the levels should be read with this in mind.
- The multi-day VaR models reuse the daily EVT marginal shock shape scaled by the h -day volatility forecast. Because temporal aggregation typically thins heavy tails toward normality, reusing daily marginal tail shapes may overstate tail heaviness at longer holding periods (such as 20 days).
- The portfolio weights are fixed and there are no transaction costs, rebalancing or liquidity effects.

6 Conclusion

Tail dependence does change with the investment horizon. The change is clear for equities and credit, likely for equities and Treasuries, and in the opposite direction for gold. A risk model that uses daily dependence for a 20-day horizon understates VaR by about 5.0% on average and by more than 10.9% on one day in ten (comparing daily model forecasts rather than realised breach losses, with an out-of-sample Diebold-Mariano test finding no statistically significant separation, $p = 0.32$). The $\sqrt{h}$ rule hides this by overstating volatility. Measuring horizon risk with full-horizon volatility and horizon-matched dependence fixes both errors, scored best out of sample, and needs less capital than the rule it replaces.

Reproducibility and use of AI tools

`python run_all.py` rebuilds every number, table and figure in this report from the stored price data in about six minutes, and gives identical output on repeated runs on a single machine (runs across different CPU architectures or BLAS libraries can differ in a final digit due to floating-point rounding). The numbers in the text are written by the same script to `tables/key_numbers.tex`.

Package versions are pinned in `requirements.txt` (Python 3.12), and `pytest` runs the test suite.

AI assistants were used for coding help, debugging and drafting text. The team chose the question, data, methods and conclusions, checked every result against the code, and takes full responsibility for the submission. Details are in `docs/AI_DISCLOSURE.md`.

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